NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

Requests shall be referred to Headquarters, U.S. Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, MD 21005-5055, usarmy.APG.atec.mbx.atec-standards@mail.mil

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REFERENCE FILE

24 September 1970

US ARMY TEST AND EVALUATION COMMAND
COMMON TROPIC ENVIRONMENTAL TEST PROCEDURE

GENERAL SUPPLIES AND EQUIPMENT

1. OBJECTIVE

This document is a guide to test methods and techniques for determining the ability of supplies and equipment to be stored and operated effectively within humid tropic environments.

2. BACKGROUND

Supplies and equipment can often be transported, stored and then operated effectively and efficiently in a temperature environment, but when similarly exposed to tropic environmental and meteorological factors suffer degradation or become inoperative. The conditions existing inside a closed package or container are not necessarily the same as the external environment. Internal package temperature may be significantly higher than the ambient temperature; if the package is sealed, external temperature variations result in changes in the pressure and relative humidity inside the package. Thus, stored items experience tropic exposure long before being made operational. High temperature and humidity increase moisture penetration into the stored or operative item(s); if salt spray is present, the high temperature increases the corrosive effects of the spray. Similarly, deterioration of rubber components is greatly increased by ozone and solar radiation in tropic temperatures. Test item subject to storage in shipping containers in the tropics must be properly resealed following periodic inspection to approximate as close as possible the original packaging condition. Finally, the preponderance of insects and propensity of fungus and microbiological growths to thrive in the tropics act to rapidly deteriorate storage containers, test item structural components, and degrade operating efficiency. Testing, therefore, is required to determine if new equipment, after having met stated performance criteria in temperate and/or simulated tropic environments, will indeed function properly and maintain its operating efficiency when shipped to and operated and/or stored in the humid tropics. (See Appendix A in MTP 2-4-003 for a detailed description of the environmental and meteorological conditions found in the tropics).

3. REQUIRED EQUIPMENT

One or more of the following items and/or facilities may be required to obtain data during the various evaluations:

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a. Facilities and measuring and recording devices listed under Paragraph 3., "REQUIRED EQUIPMENT", in the specific MTP for the item under evaluation.
b. Photographic equipment (still and motion).
c. Appropriate material handling equipment (HME).
d. Platform-type weighing scales.
e. Meteorological equipment.
f. Item under test (test item).
g. Suitable test sites and storage areas as defined in MTP 2-4-003.
h. Reference or comparison item(s) as required.
i. Camouflage equipment as required.

4. REFERENCES

A. AR 70-38 Research and Development: Research, Development, Test and Evaluation of Materiel for Extreme Climatic Conditions.
B. USAMC Regulation 385-12 Safety: Verification of Safety of Materiel from Development Through Testing, Production, and Supply to Disposition.
C. USAMC Regulation 385-224 Safety: Safety Manual.
D. USAMC Pamphlet 702-3 Quality Assurance: Reliability Handbook.
E. USAMC Pamphlet 706-134 Engineering Design Handbook: Maintainability Guide for Design.
F. USATECOM Regulation 70-23 Research and Development of Materiel: Equipment Performance Reports (EPRs).
G. USATECOM Regulation 385-6 Safety: Verification of Safety of Materiel During Testing.
H. USATECOM Regulation 700-1 Quality Assurance: Value Engineering.
I. USATECOM Regulation 750-15 Maintenance of Supplies and Equipment: Maintenance Evaluation During Testing.
J. USAGETA Document Human Factors Evaluation Data for General Equipment (HEDGE) Guidebook Supplement.
K. MIL-STD-129 Marking for Shipment and Storage.
L. MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of.
M. MIL-STD-1472 Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
N. TM 743-200 Storage and Materials Handling.
O. FM 31-30 Jungle Training and Operations.
P. MTP 2-4-003 Tropic Environmental Test of Wheeled and Tracked Vehicles.
Q. MTP 8-4-004 Long Term Surveillance/Environmental Testing of CBR Munitions, Weapons.

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R. Specific Volume 10 MTP for item and/or category under evaluation.

5. SCOPE

This MTP is intended to be used as a guide in conjunction with the MTP for the specific commodity item(s) undergoing service tests. Reference is made to other MTPs for the actual test procedure to be followed in assessing the test items physical condition, technical characteristics, and operating performance. The primary intention of this MTP is to specify the conditions under which such testing is to be performed, collect and analyze data to determine the suitability of the test item for use and storage in the humid tropics.

5.1 SUMMARY

This MTP describes the following tests to be performed in a humid tropic environment on general supplies and equipment.

5.1.1 Preparation for Test

This section provides guidance for test project planning, a discussion of facilities and equipment required, instructions for test personnel training and familiarization, and a typical mission scenario.

5.1.2 Test Conduct

a. Physical and operating characteristics - A determination of the test item condition upon its arrival and an assessment of its operating characteristics prior to conducting environmental testing. This data will be used as a basis for comparison with corresponding data developed during subsequent testing.

b. Operational Performance - An evaluation of the suitability of mechanical and electro-mechanical items for continued use in the humid tropics.

c. Individual Equipment - An evaluation of the suitability of equipment worn, carried, and used by the individual soldier in the humid tropics.

d. Tents and Shelters - An evaluation to determine the suitability of the tents and shelters to be repeatedly stored, transported and used in the humid tropics.

e. Clothing - An evaluation to determine fit, comfort and wear characteristics under both combat and garrison conditions; and in addition, the protection afforded the wearer from the humid tropic environment.

f. Armor and Related Equipment - An evaluation to determine fit, wear, comfort, and protection provided when stored and used in a humid tropic environment.

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g. Foodstuffs - An evaluation to determine the palatability and resistance to tropic environmental influence of packaged foodstuffs when transported and consumed in the humid tropic environment.

h. Short Term Storage - An evaluation of the ability of the test item(s) and/or components to be stored for short intervals in the humid tropics and not suffer loss of operability, efficiency, structural strength or integrity, or safety in operation.

i. Surveillance - An evaluation of the ability of the test item to withstand long term storage, while properly packed and packaged for storage in a humid tropic environment.

j. Security from Detection - An evaluation to determine the capability of the test item to avoid detection when stored or used in a humid tropic environment.

k. Maintenance Evaluation - An evaluation to determine the maintenance characteristics of the test item.

l. Safety - An evaluation to determine the safety characteristics and possible hazards to the test item user with specific emphasis on those characteristics peculiar to the humid tropic environment.

m. Human Factors - An evaluation to determine the suitability in humid tropic environment of those design and performance characteristics affecting the test item user.

n. Value Analysis - An evaluation directed at analyzing the primary function and features of the test item for the purpose of cost reduction without compromising performance reliability, quality, maintainability or safety.

5.1.3 Test Data

This section details the raw data to be collected and recorded while completing the test procedures in paragraphs 6.2, Test Conduct.

5.1.4 Data Reduction and Presentation

This section provides instructions for analyzing and evaluating the raw data and presenting the results.

5.2 LIMITATIONS

a. The material presented in this MTP is limited to field testing in the humid tropics. Guidance for testing equipment in simulated environments or other induced tropic conditions has been intentionally omitted.

b. Procedures outlined in this MTP do not constitute detailed test plans. Humid tropic environment testing should be performed utilizing the guidance in this MTP, but each cognizant agency must determine the applicability of the procedures and how best to obtain the required data from the test item(s).

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6. PROCEDURES

6.1 PREPARATION FOR TEST

6.1.1 Test Project Planning

The test project officer and other designated personnel must:

a. Conduct a thorough study of stated requirements as contained in QMRs, SDRs, TCs, and the test directive to insure that complete and suitable test criteria are selected.

b. Review all instructional material issued with the test item by the manufacturer and reports of previous tests conducted on the item or similar equipment.

c. Prepare a detailed test schedule showing proposed time periods allotted for each test. As applicable, schedule testing in both cleared and uncleared terrain in wet-warm and wet-hot environments. These environments and test courses are defined in paragraph 3 and Appendix A of MTP 2-4-003. Testing should be scheduled in these areas during the peak of the wet season (October through November) and the dry season (January through March). Wherever applicable, testing should also be conducted in areas that cause salt laden air and salt spray exposure to the test item(s). The tropic environment is most likely to offer severe degradation effects to test items emplaced in the wet season followed by the dry season in lieu of the reverse condition.

d. Prepare record forms and test logs for systematic entry of data, chronology of test, and evaluation in the final analysis for the test item's suitability for use in the humid tropics and whenever applicable, prepare an annex to the USATTC local safety SOP when specific safety procedures are undefined.

e. Review the safety release. Tabulate the safety precautions that must be followed to ensure safe operation of the test item(s) include all applicable safety SOPs.

f. Plan the utilization of photographic techniques where necessary to record and document test results.

6.1.2 Test Personnel Training and Familiarization

a. Evaluate the adequacy of all draft technical manuals and safety instructions used by the test personnel. This evaluation shall continue throughout the course of testing.

b. Instruct personnel as follows:

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Table 1. Personnel Instructions

Test Participants	To be instructed in
Testing personnel	a. The test or reference item operating instructions. b. The characteristics of the reference item, if any. c. The kind of data to be obtained. d. The environmental and meteorological conditions in which testing is to be conducted. e. The health precautions to be observed.
Test Participants	a. The test item (and reference item, when used) operating characteristics and limitations. b. The safety precautions to be observed. c. The kind and extent of all maintenance actions to be taken under all specified environmental conditions. d. All operating procedures to be followed under all environmental and meteorological conditions. e. The procedures to be followed when filling out checklists and questionnaires, and their purpose.
Test evaluators	a. The purpose and use of all checklists and questionnaires, and the methods to be used in their evaluation. b. The mathematical models to be used in evaluating the test data, and the calculations to be made.
Topographical analysis support	a. Physical terrain features to be measured, including water velocity and terrain slopes. b. Vegetation features to be noted and described.
Maintenance support	a. The scheduled maintenance requirements to be met. b. The procedures, equipment and material to be used to make emergency repairs and unscheduled maintenance. c. The recovery procedures to be used. d. The maintenance records to be maintained.

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Table 1 (Continued)

General support	a. Storage, handling, layout, spacing, pest control, fire protection, security, and inspection procedures to be followed. b. Required organizational maintenance to be performed, including cleaning, and microbiological inspection. c. The calendar dates of all inspections.
Meteorological support (TTC Reg 705-4)	a. The terrain areas which will be used. b. The calendar dates of all tests. c. The readings to be taken.

6.1.3 User Test

6.1.3.1 General

The primary way of service testing general supplies and equipment is to perform a receipt inspection, and then issue the equipment to personnel for the period of test, to be used during training and field operations. The test officer will inspect equipment at scheduled intervals; inspect failures as they occur; interview personnel; and administer questionnaires. The procedures provided in paragraph 6.2 should be used in support of this activity. If, however, normal training and field operations are not defined, a controlled test environment will be developed and used to evaluate the test equipment as discussed in 6.1.3.2, below.

6.1.3.2 Mission Scenario

a. The general supplies and equipment will be evaluated by conducting tests under simulated field conditions by personnel who would use the equipment in the field. The mission scenario (Appendix A) describes the test areas and operations that will be used for evaluating the test item.

b. Prepare any additional mission scenarios required to include, as a minimum, the following:

1) Details of test item prime function(s) to be completed in each condition of environment and terrain.
2) Sketch map or photomap and description of expected environment to be found in each test course.

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3) Personnel required by grade and MOS to complete mission(s) planned.
4) Test equipment required to support mission(s).

c. When test items are evaluated during tactical training exercises, the test item use will be documented from the operation orders, scenarios used in conjunction with training problems, situation outlines and Army Training Tests.

6.2 TEST CONDUCT

NOTE: Conduct all testing in accordance with the applicable guidance supplied in MTP 2-4-003, paragraph 6.2.

6.2.1 Physical and Operating Characteristics

NOTE: The physical and operating characteristics of each test item must be determined prior to conducting environmental tests. This information shall be used as a base for comparison with subsequent data developed in completing the procedures of this MTP for two reasons:

1) To avoid attributing performance degradation or malfunctions to humid tropic testing when the condition was present upon test item receipt.
2) To precisely determine the amount of performance or structural degradation suffered by the test item from completing the test procedures.

a. Inspect the packaging and identification markings of items received for conformance to MIL-STD-794 for overseas shipment for level A or B, as applicable.

b. Unpack and remove all preservative coatings on test item(s). Inspect all surfaces for proper painting and other protective coatings. Take color photographs of all surfaces displaying roughness, corrosion or microbiological growths.

c. Verify completeness of the test item, associated components, and maintenance test package as specified in the Basic Issue Item List (BIIL) and file an Equipment Performance Report for any damages or omissions, if required.

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d. Determine the physical characteristics of the test item(s) in accordance with the appropriate procedures in the applicable volume 10 MTP. Prepare an EPR for any damages detected.

e. Using the procedures in the applicable draft technical manuals, prepare the test item for normal operation; complete all functions of the test item in all normal modes and ranges of operation.

6.2.2 Operational Performance

The following procedures are provided to determine the suitability of all volume 10 equipment for use in the tropics except those commodity items specifically described in subsequent paragraphs. (These being foodstuff, personal equipment, clothing, tents and shelters). Where an item is not functionally complete, it should be tested while used in the manner for which it was intended. For example, a hose will be installed in a hydraulic system and the whole system shall be operated and tested as a unit.

a. Using instructions in the draft technical manual, prepare the test item for operation in the designated test site.

b. If the test item(s) has not previously been used, perform specified break-in actions.

c. As described in the applicable Volume 10 MTP, perform all functions for which test item was intended to determine if these functions can be successfully completed in the test environment.

d. Continue test operations for sufficient number of missions to determine the suitability of the test item for continued operation in the test environment.

NOTE: Insure that all preventative maintenance, cleaning and lubrication actions prescribed by the draft technical manual for operation in the test environment are followed.

e. Repear steps a. through d. for each test site and environmental condition selected for testing as described in paragraph 6.1.1.

6.2.3 Individual Equipment

The items in this category consist of individual equipment, such as messing equipment and all items and tools worn, carried and used by soldiers. Not included in this category are tents, clothing and related armor.

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a. Prepare the test item for use, and complete all functions for which it was intended.

b. Clean and prepare the test item for transport (by manpack, if possible) and transport the item in a typical mission. Expose the transported item to all typical field abuses (exposure to weather, immersion while fording streams, carrier induced shock, etc).

c. Repeat foregoing steps for sufficient time to determine if the individual equipment remains serviceable and maintains its appearance when used in the humid tropics.

6.2.4 Tents and Shelters

a. Erect the tent or shelter.

b. Disassemble and prepare the tent or shelter for transport (by manpack, if possible).

c. Transport the item in a typical mission.

d. Erect the tent or shelter and allow it to be exposed to rain, extreme heat and humidity, direct sunlight, extreme heat and dryness, for sufficient time to evaluate their effect and the effect of insects and microbiological growths on the test item.

e. While completing step d., perform the following blackout test:

1) Select a moonless night for testing.
2) Prepare the tent or shelter for blackout testing by covering all entrance/exitways and windows with the flaps provided or other specified blackout material.
3) Activate lamps normally expected to be used in the tent or shelters during an actual blackout.
4) Circumnavigate the tent or shelter at sufficient distance so as to not detect any light emanating from it. Close in on the test item while orbiting about it. Note the maximum distance and associated azimuth at which light is visible. Note the location on the tent where light is visible. Determine if light is visible at any other azimuthal position.
5) Attempt to improve blackout material and procedures such that radius of light detection is minimized. Repear step 4) and this step until no further improvement is practical.
6) Repear steps 4) and 5) while orbiting the tent or shelter from the air. Determine minimum orbital distance at which effective blackout can be maintained.

f. Repeat steps b., c., d., and e. until tent or shelter is no longer serviceable or expiration of test period occurs.

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6.2.5 Clothing

Refer to and comply with the requirements of the applicable Volume 10 MTPs on clothing in performing the following:

a. Prepare clothing for use and complete test preparation as described in the associated MTP.
b. Don clothing, noting any difficulties, presence and adequacy of adjusting and fastening devices, odor, and time required to completing donning operation.
c. Perform basic motions (bending, stretching, twisting, sitting) to determine the comfort and degree of body freedom allowed by the clothing.
d. Determine doffing time for normal and emergency operations.
e. Complete test missions as described in the applicable MTP to determine if the test item can be comfortably worn during extended periods, and provided the protection from the humid tropic environment specified.
f. Complete field laundering operations to determine the ease/difficulty in maintaining specified levels of cleanliness.
g. Cycle test item through standard launder, storage, mission operation procedures to determine the wear characteristics, odor, resistance to color fade, resistance to shrinkage, and resistance to deterioration from insects.

6.2.6 Armor and Related Equipment

Refer to and comply with the requirements of the applicable Volume 10 MTPs on armor and related equipment in performing the following:

a. Prepare test item for use and complete test preparation as described in the associated MTP.
b. Don test item noting any difficulties, presence and adequacy of adjusting and fastening devices, odor, and time required to complete donning operation.
c. Perform basic motions (bending, stretching, twisting, sitting) to determine the comfort and degree of body freedom allowed by the test item.
d. Determine doffing time for normal and emergency operations.
e. Complete test missions as described in the applicable MTP to determine if the test item can be comfortably worn during extended periods, and provide the protection specified.
f. Complete field and base laundering operations to determine the ease/difficulty in maintaining specified levels of cleanliness.

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g. Store the item in the manner normally expected to be employed between missions. At the completion of the storage period examine the item for evidence of color fade, deterioration from insects, corrosion, or microbiological growths. Repeat test actions specified in step e.

6.2.7 Foodstuffs

NOTE: Wherever possible, conduct the following procedures in conjunction with the field training of units stationed in the Zone.

a. Plan a mission as described in Appendix A to include traversing streams and personnel exposure to varied weather.

b. Prepare the food items under test in support of this mission for backpack transport.

c. Complete the planned mission. At completion of each period, unpackage a portion of the food items and prepare the meals required.

d. Continue transport and consumption of food items in the manner just described until the mission is completed or a determination is made that the food, its packaging or both is not suitable for use in the humid tropics.

e. Make use of photography to illustrate initial condition of foodstuff and any degradation of food or packing detected.

6.2.8 Short Term Storage

Storage testing is conducted to determine if the test item(s) can withstand exposure to humid tropic environments for brief periods and with a minimum of protection and not suffer structural or performance degradation. Of particular interest is the effects of short term storage on textile goods (such as clothing and parachutes), leather goods, personnel armor, foodstuffs and packaging of foodstuffs. The storage tests are to be conducted as follows, consistent with normal practice for the commodity class of the test item as supplemented by the information in governing documents:

a. Place the test item and/or components in the area designated for short term storage. Unless otherwiese specified, minimum use of preservatives should be made.

b. Periodically inspect the stored items for evidence of corrosion, microbiological growths, insect and rodent damage, and failure of test item packaging. Sample and identify any growths detected.

c. At completion of the storage period, repeat applicable procedures of foregoing sections to determine any performance or structural degradation resulting from storage testing. In addition, complete the following evaluations:

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1) For foodstuffs:

a) Change of color, odor, texture.
b) Deterioration of packaging.
c) Infestation of insects.
d) Loss of palatability.

2) For textiles:

a) Deterioration of fabrics and/or connective material resulting from mildew, insects, etc.
b) Instances of shrinkage or fading.
c) Increased donning and/or doffing time.
d) Adverse psychological reactions of wearer.
e) Inability to safely and effectively perform prime function.

3) For leather goods:

a) Loss of appearance or strength.
b) Instances of cracking or enbrittlement.
c) Adverse psychological reactions of user.
d) Inability to satisfy performance requirements.

4) For personnel armor.

a) Loss of strength and ability to satisfy protection requirements.
b) Loss of appearance
c) Increased donning and/or doffing time.
d) Adverse psychological reactions of wearer.

6.2.9 Surveillance

Surveillance testing is performed to determine the effect of long term storage on materials and/or performance. These items are stored and tested on a continuing basis in the tropic environment to determine the extent of deterioration as compared with original requirements.

a. Prepare the test item for long-term storage in the tropics as described in the draft technical manual or other governing documents.

b. Conduct surveillance testing in accordance with the applicable guidance supplied in MTP 8-4-004. As described therein, complete periodic inspections and perform the applicable procedures of foregoing sections to determine any performance or structural degradation resulting

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from long-term storage. Sample and identify any microbiological growths detected.

6.2.10 Security from Detection

NOTE: This subtest should be conducted when specifically requested in the test directive.

a. Select as many forms of camouflage as would be used in field conditions. A trailer mounted repair shop, for example, may be camouflaged in shadows of buildings, placed under trees, placed under artificial canopy hung between trees or poles, placed in a depression and covered by nets, or painted to blend in and match the general terrain. For items provided already containing a method of camouflage, assess the concealment qualities offered and determine if any additional camouflage material is required.

b. For items that are to be inoperative while concealed, place in designated site and use camouflage and camouflage forms selected to conceal test item. Place material about site as required to enhance concealment qualities. For items containing camouflage, testing should also be completed with no additional camouflage material employed.

c. For personal camouflaged equipment, items such as body armor, clothing, helmets, etc., conduct a tactical operation with personnel wearing the test items. Personnel should proceed as if engaging hostile troops, making use of terrain features to enhance concealment qualities.

d. Make ground and aerial observations of test item(s) and control items (and activity) at various times of the day and at various distances from the item(s) (and activity) to assess concealment achieved.

e. Repeat foregoing steps in each test site of interest.

6.2.11 Maintenance Evaluation

Data expressing system maintenance factors are collected in preparing for and completing active testing. If at the completion of testing very little maintenance has been performed, representative samples of the maintenance procedures in the applicable Volume 10 MTP and draft technical manual should be completed to determine the ease in performing these actions with emphasis on the following:

1) Organizational (O), Direct Support (F), and General Support (H) maintenance requirements.
2) Operator through General Support Maintenance Literature.
3) Repair parts.
4) Tools.
5) Test and handling equipment.
6) Calibration and maintenance facilities.
7) Personnel skill requirements.

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6.2.12 Safety

a. Do not begin testing until a safety release is received from USATECOM.

b. Observe the proper safety precautions during testing, noting any conditions that might present a safety hazard.

c. Observe procedures and complete safety evaluations as specified in the following documents.

1) The test item maintenance test package.
2) USATECOM Regulation 385-6.
3) FM 31-30.
4) Applicable portions of Appendix B in MTP 2-4-003 (Questionnaires).
5) Applicable portions of Appendix C in MTP 2-4-003 (Checklists).
6) Applicable portions of specific commodity Volume 10 MTP.

d. Complete the following additional safety evaluations.

1) Presence and clarity of safety warnings on test item.
2) Adequacy of handling and operating procedures with regard to personnel safety.
3) Presence, sufficiency, and adequacy of safety devices.
4) Presence of sharp or projecting edges, controls or other hardware.
5) Accessibility to emergency cut-off controls.
6) Adequacy of emergency equipment and instructions, if applicable.
7) Adequacy of fire fighting equipment and fire emergency procedures.

6.2.13 Human Factors Evaluation

Develop task/item checklists reflecting the human factors design considerations for the test item. These checklists will allow test supervisory personnel to compare test item features against appropriate criteria and record comments to evaluate the suitability of the man item interface with particular emphasis on operations performed under adverse weather conditions. Detailed criteria and human factors considerations for each task may be derived from USAGETA document "Human Factors Evaluation Data for General Equipment (HEDGE) Guidebook Supplement." Consult Appendix C in MTP 2-4-003, MIL-STD-1472, and the following for guidance in preparing a suitable checklist.

a. For all tasks the following factors will be considered:

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1) Adequacy of instructions and tools to perform the task.
2) Mental and physical effort required.
3) Design of the test item as it affects the task.
4) Time required for the task.
5) Personnel required for the task.
6) Compatibility with other equipment.

b. Perform the following tasks for the HEDGE test functions given. The factors considered shall include, but not be limited to, those of section a. above.

1) Operability.

a) Assemble and set-up.

1. Assemble components and make all connections.
2. Make preliminary alignment, calibrate and adjust.

b) Prepare

1. Check controls, fasteners, connectors.
2. Load expendables (lubricants, fuel, etc.)
3. Close covers, caps, etc.

c) Operate test item.
d) Energize engine.

1. Operate-manipulate controls; perform prime function(s).

2) Maintainability.

a) Perform scheduled maintenance

1. Clean, add lubricants.
2. Remove and replace minor items.
3. Tighten fasteners, connectors.
4. Adjust, calibrate, align.

b) Perform nonscheduled maintenance.

1. Detection of malfunction by observing displays, noting visual or audible changes, or changes in operating effectiveness.

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2. Isolate and identify causes by visual means or instrumentation.

c) Remove and replace.

1. Open, gain access to and remove component.
2. Replace or repair and re-establish proper operation.

3) Transportability

a) Prepare for transport.

1. Place in the transit state by removals, collapsing, tightening, locking, applying protective cover.
2. Package/containerize, block/brace the test item accessories.

b) Load/unload.

1. Move test item to the carrier; use appropriate MHE and move test item accessories to the carrier.
2. Place test item and accessories into/onto the carrier.

c) Secure/unfasten.

1. Tie down or secure
2. Remove test item from carrier.
3. Open container, unblock, remove and reinstall test item accessories.

6.2.14 Value Analysis

a. During the conduct of all tests, test personnel shall evaluate the test item from a value versus cost standpoint noting features or components which can be eliminated or modified to accomplish cost reduction without impairment of performance, reliability, quality, maintainability, or safety. The applicable portions of USATECOM Regulation 700-1 shall be used for this evaluation.

b. Consideration shall be given to the topics listed below:

1) Mission Capacity

The test item should be capable of accomplishing the specified task with only a reasonable margin of excess

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capability. Excess capacity and unused capability normally result in unnecessary bulk, excessive weight and unwarranted costs.

2) Simplicity

Unnecessarily complex components and systems, redundancy, and the use of unneeded parts will increase costs and maintenance efforts.

3) State of the Art

In may instances the use of recently developed, currently available, components and automated features will result in an overall product improvement and cost savings.

4) Standardization

The use of identical parts and parts currently in the military system will reduce the overall logistics burden.

5) Materials and Methods of Construction

Polished surfaces, overdone finishes, and the use of expensive materials will result in unnecessary costs if used inappropriately.

6) Tolerances

Inadequate tolerances will result in difficulties and delays in accomplishing post arrival assembly, routine maintenance, servicing and repair.

6.3 TEST DATA

NOTE: Refer to Appendix B for general checklist and tabulation of environmental effects on test item operation in the humid tropics.

a. When measuring attributes, which are subject to small deviations, make at least two, and preferably four, different measurements under identical test conditions, and record each measurement, as well as the arithmetic mean of these measurements.

b. Indicate the accuracy (and calibration data, when applicable) of the measuring device employed when recording measurements which must be made accurately.

c. When progressive degradation is observed on any part, describe and/or photograph the degradation, and show the "before" and "after" condition together when recording.

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d. When a group of people are asked to fill out a questionnaire giving their opinion regarding specified features in a particular test, show the scoring values used, and the scoring results obtained, in tabular form as described in Appendix B of MTP 2-4-003 (Questionnaires).

e. When evaluating test item degradation resulting from humid tropic environment testing, ensure, as a minimum, that the following indications of deterioration and appropriate meteorological data are recorded as they apply.

1) Loss of Calibration.
2) Destruction of thermal devices.
3) Evaporation or flow of lubricants.
4) Malfunction of electrical contacts due to heat expansion.
5) Bind of mechanical parts due to differential expansion of dissimilar metals or to loss of lubricants.
6) Permanent set of packings and gaskets.
7) Adhesion of closure and sealing strips (due to partial melting) to contacting parts.
8) Discolor, crack, bulge, check, craze, or strength reduction of materials.
9) Corrosion.
10) Crystallization of rubber components.
11) Microbiological growths, particularly on the following:

a) Textiles.
b) Wood and paper.
c) Foodstuffs.
d) Rope and cordage.
e) Leather.
f) Cork.
g) Gaskets (fiber, felt, cork).
h) Electrical insulating material.
i) Adhesives.
j) Plastics and plasticizers.
k) Vinyl coverings.
l) Paints and varnishes
m) Synthetic resins.
n) Sealing compounds
o) Lens coatings and cements.
p) Glass (through etching).
q) Explosives and propellants.
r) Lubricants.
s) Rubber.
t) Secondary damage through decomposition of protective coatings or seals, thus admitting moisture leading to corrosion or deterioration of electronic components

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or instruments that were supposed to be hermetically sealed.

12) Moisture absorption, swelling, chemical or organic decomposition, saturation of protective desiccants.
13) Electrical or electronic difficulties.

6.3.1 Test Personnel Training and Familiarization

Record the following:

a. For all test personnel:

1) MOS.
2) Rank.
3) Unit.
4) Experience.
5) Previous training and skill level.

b. Adequacy of draft technical manual for supporting personnel training.
c. Requirements for training aids.
d. Difficulties encountered in completing training program.

6.3.2 Physical and Operating Characteristics

Record the following:

a. Conformance with specified packing, packaging, preserving, identification and marking criteria.
b. Any omissions of components or equipment damage detected.
c. Physical characteristics of the test item as specified in the applicable Volume 10 MTP.
d. Instances of missing or improper protective coatings on test item. Identity of photographs illustrating roughness, corrosion, mildew or other microbiological growths.
e. Description of operational condition of test item. Provide enough details of test performed and results obtained to evaluate the operating efficiency of the item in all modes, ranges, and speeds of operation, as applicable.

6.3.3 Operational Performance

Record the following for each test site, environment and season (wet, dry) in which testing was conducted.

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a. Identity of test item.
b. Data required by applicable Volume 10 MTP detailing the operational performance achieved.
c. Instances wherein pecularities of test site, environment or season impeded test operations or reduced operating efficiency of test item. provide sufficient details of test conduct and resulting data to accurately determine what components were affected and the extent to which functions were impaired.
d. Any deterioration of test item performance or structural conditions resulting from environmental testing. Include particulars as to elapsed time, meteorological conditions during test, type of fungus and amount of corrosion detected, projected shortened life span of components affected, etc.

6.3.4 Individual Equipment

Record the following for each test site, environment and season (wet, dry) in which testing was conducted.

a. Identity of test item.
b. Data required by applicable Volume 10 MTP detailing the operational performance achieved.
c. Description of test actions completed; and reduction in operatability, accuracy, or appearance of test item so caused. Include particulars as to elapsed time of testing, meteorological conditions during test, type of fungus and amount of corrosion detected, etc.

6.3.5 Tents and Shelters

Record the following:

a. Identity of test item.
b. Ease/difficulty in transporting test item; any damage so caused.
c. Ease/difficulty in erecting tent or shelter; requirements for manpower and time.
d. Suitability for use in each environment tested. Include particulars as to amount of moisture penetration, cracking and separation of fabrics, effect of insects and microbiological growths, and any other durability indicators observed.
e. Number of test cycles (transport and erection of tent or shelter) completed; elapsed time of each test cycle.
f. Description of ambient lighting conditions under which blackout testing was conducted.

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g. Description of blackout procedures followed, blackout material and light sources used during blackout testing. For each light source include intensity rating and relative location inside test item with respect to doors and windows.

h. Improvements, if any, made to blackout procedures and materials specified to be used.

i. Radius and corresponding azimuth(s) at which blackout is ineffectual on ground and air. For aerial observations include also elevation from horizontal.

6.3.6 Clothing

Record the data required by the applicable MTP and the following:

a. Identity of test item.

b. Ease and time involved in donning and doffing clothes in both normal and emergency situations.

c. Ease in using fastening and adjusting mechanisms; suitability in number and range of adjustment devices provided.

d. Comfort of clothing.

e. Compatibility of clothing with standard field items and equipment.

f. Agreement/disagreement of protection offered wearer with stated requirements.

g. Ease/difficulty in completing field laundering operations. Degree of cleanliness maintained.

h. Effects of standard launder process on test items. Suggestions for change of process.

i. Average shrinkage encountered; loss of coloring.

j. Average durability achieved for various items (number of cleanings, number of times worn, total wear time).

k. Appearance of test item upon receipt and throughout test.

l. Brief description of test conduct and conditions under which evaluations completed.

6.3.7 Armor and Related Equipment

a. Identity of test item.

b. Ease and time involved in donning item in both normal and emergency situations.

c. Ease in using fastening and adjusting devices; suitability in number and range of adjustment devices provided.

d. Compatibility of test item with standard field items and equipment.

e. Comfort and degree of body freedom provided by test item.

f. Agreement/disagreement of protection offered wearer with stated requirements.

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g. Ease/difficulty in completing field and base laundering operations. Degree of cleanliness maintained.

h. Results of storage and subsequent testing.

6.3.8 Foodstuffs

Record the following:

a. Identity of foodstuff, type packing and illustrative photography.

b. Any deterioration of food packaging; conditions of test under which deterioration occurred, including elapsed time prior to inception of deterioration.

c. Any disagreeable conditions of foodstuffs resulting from test actions, including a description of the actions performed and elapsed time of active testing.

1) Change of color, odor, texture.
2) Infestation of insects.
3) Loss of palatability.

6.3.9 Short Term Storage

Record the following:

a. Identity of test item.

b. Condition of the test item(s) just prior to storage, and the storage preparations made, including (if applicable) the shipping containers used.

c. Storage environment and average meteorological conditions.

d. Number of items involved.

e. Method and duration of storage of each item.

f. Results of inspections made.

g. Maintenance and repairs made.

h. Final condition of test item.

6.3.10 Surveillance

Record the following:

a. Identity of test item.

b. The condition of the test items just prior to storage, and the storage preparations made, including all containers and packaging used.

c. Storage conditions and average meteorological conditions.

d. Number of items involved.

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e. Method and duration of storage of each item and the calendar dates of all inspections.
f. Results of all inspections made.
g. Maintenance and repairs made.
h. Final condition of test item.

6.3.11 Security from Detection

Record the following for each camouflage condition:

a. Identity of test item; forms of camouflage used.
b. Concealment qualities offered by each form of camouflage; concealment offered by camouflage agents previously applied to test item, and any requirements for additional camouflage material; requirements for natural and/or man-made landmarks/terrain features to complete concealment.
c. Description of test conduct to include the following:

1) Characteristics of terrain in which observations were made.
2) Implements used to make observations (field glasses, aircraft, etc.).
3) Distances at which observations made.
4) Time of day and sky conditions when observations were made.

d. Time required and difficulty to complete camouflage actions.
e. Any loss of camouflage qualities of clothing due to launder and aging effects.

6.3.12 Maintenance Evaluation

Record the following:

a. Scheduled maintenance:

1) Description of test activity, environment, and meteorological factors associated with maintenance performed.
2) Procedures and special tools (if any) required to complete maintenance actions; description of maintenance performed.
3) Component or feature involved.
4) Elapsed time since previous scheduled or unscheduled maintenance performed (whichever is shorter). Identity of previous maintenance performed (scheduled or unscheduled).

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5) Parts required, if any.
6) Elapsed time required to complete maintenance actions.
7) Difficulties in completing maintenance actions.
8) Total scheduled maintenance time consumed in hours.

b. Unscheduled maintenance:

1) Description of test conduct, environment, and meteorological conditions under which maintenance requirements developed.
2) Procedures and special tools (if any) required to complete maintenance actions; description of maintenance performed.
3) Elapsed time since previous scheduled or unscheduled maintenance performed (whichever is shorter). Identity of previous maintenance performed.
4) Parts required, if any.
5) Elapsed time required to complete maintenance actions.
6) Difficulties in completing maintenance actions.
7) Component of features involved and method used to determine it.
8) Damage caused to associated parts of the system by failure, if applicable.
9) Reason for maintenance required, if ascertainable.
10) Recommendations to prevent recurrence of maintenance requirements.
11) Total unscheduled maintenance time consumed in hours.

c. Time to complete related maintenance functions.

1) Supply delay time.
2) Administrative delay time.
3) Technical engineering investigations or inspections related to analysis of cause of failure or detection of suspected incipient failures.
4) Initial and final technical inspections.

d. For all maintenance performed:

1) The adequacy of the draft technical manual and maintenance information provided.
2) The adequacy of the OEM tools and repair parts.
3) The adequacy of the safety instructions.

e. Total operating or in-use time in hours.

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f. Identify each maintenance action as either a chargeable or nonchargeable failure in accordance with definitions and illustrations provided in USATECOM Regulation 750-15.

1) Total number of chargeable failures.
2) Total number of nonchargeable failures.

6.3.3 Safety

Record the appropriate data as required by the specific Volume 10 MTP for the test item and the following:

a. A tabulation of all safety features and/or devices.

1) Type of feature/device.
2) Purpose of the feature/device.
3) Suitability of the feature/device.
4) Adequacy of the feature/device.
5) Proper operation of the feature/device.

b. Adequacy of warning plates, instructions and markings in content, clarity, sufficiency and location.

c. Any condition that did or might present a safety hazard including cause and corrective action required to alleviate the condition.

d. Presence and adequacy of fire fighting equipment and suitability of stowage and control location.

e. Adequacy of ventilating systems to prevent operator exposure to noxious fumes.

f. Satisfactory/unsatisfactory noise levels during testing.

g. Suggestions to improve safety features, safety measures and/or precautions.

6.3.4 Human Factors Evaluation

Complete the task/item checklists by rating the inclusion of each design consideration as satisfactory or unsatisfactory.

6.3.5 Value Analysis

a. Record appropriate comments for each of the topics listed below:

1) Mission Capacity.
2) Simplicity.
3) State of the Art.
4) Standardization.

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5) Materials and Methods of Construction.
6) Tolerances.

b. When making recommendations for changes in test item features or components, record the following:

1) The feature or component under consideration.
2) Recommended change(s).
3) Reason(s) for recommended change(s).

6.4 DATA REDUCTION

6.4.1 General

Summarize all data using tabulations and/or charts as appropriate. Analyze and compare the data collected against specific criteria stated in governing documents. Provide a narrative description of the degree of suitability of the test item for use in the tropics. If the item is not suitable for use in the tropics, provide a complete description (including test results) of why the item is so adjudged.

6.4.2 Questionnaires

When a group of people are asked to fill out a questionnaire giving their opinion regarding specified features in a particular test, and an analysis of variance is used to test for the significance of the variation in the scoring results, perform this analysis as specified in Appendix B of MTP 2-4-003.

6.4.3 Checklists

Group the answers to all checklist questions under the applicable headings, such as safety, accessibility, and human factors, and arrange the answers in the order of their relative importance.

6.4.4 Maintenance Evaluation

Calculate the maintenance indicators as follows: Reference is made to USATECOM Regulation 750-15 for information as to the meaning and use of these indicators in assessing system maintenance.

a. Mean Time Between Failures (MTBF)

$$\text{MTBF} = \frac{\text{Total operating time (in hours)}}{\text{Total number of chargeable system failures}}$$

b. Mean Time Between Maintenance (MTBM)

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$$MTBM = \frac{\text{Total operating time (in hours)}}{\text{Total number of scheduled and unscheduled maintenance actions performed}}$$

c. Mean Active Maintenance Downtime ($\bar{M}$)

$$\bar{M} = \frac{\text{Total scheduled and unscheduled maintenance time (in hours)}}{\text{Total number of scheduled and unscheduled maintenance actions}}$$

d. Mean Time to Repair (MTTR)

$$MTTR = \frac{\text{Total unscheduled maintenance time expended on chargeable failures (in hours)}}{\text{Total number of chargeable system failures}}$$

e. Inherent Availability (A_i)

$$A_i = \frac{MTBF}{MTBF + MTTR}$$

f. Achieved Availability (A_a)

$$A_a = \frac{MTBM}{MTBM + \bar{M}}$$

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APPENDIX A

Sample Scenario

Location: US Army Tropic Test Center, Fort Clayton, Canal Zone ($79^{o}57'30"W, 9^{o}21'30"N$)

Exposure: Tropical Rain Forest

A. GENERAL

1. Ten USARSO infantry soldiers will be used to conduct a reconnaissance patrol. The test officer will be assigned duties as the patrol leader.

2. For realism, six aggressors will be used to fire at the patrol causing the patrol members to deploy and maneuver (three separate actions).

3. The patrol route selected will cause the patrol to move through various types of jungle growth and terrain.

4. Blank ammunition will be issued to the patrol members and aggressors.

5. This scenario is applicable to items of clothing, individual load carrying equipment, body and head armor, individual foods and other items required to sustain the combat soldier.

B. VERBAL OPERATIONS ORDER: (The following is a verbal patrol order that will be issued at the line of departure. This order will be issued by the test officer, acting as the patrol leader. He has just returned from a briefing at S-2).

1. SITUATION

a. Enemy Forces:

1) The 2nd Battalion of the 135th Aggressor Regiment has been reported in our area of operation. They reportedly crossed the river last night, returning from R&R in a secluded area. The Battalion is estimated to be at approximately 95 percent strength and morale is high. The entire Battalion, or a portion of the Bn is capable

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of attacking our base camp and/or routes of communication. His intention is unknown.

2) No change in the status of local aggressor groups. They are capable of hit and run type tactics.

b. Friendly Forces:

1) The Bn will remain in its present defensive position and conduct extensive patrolling to eliminate the possibility of a surprise attack.

2) The Bn Mortar Platoon will be in direct support of our patrol. They will monitor our radio frequency. Calls for fire will be sent directly to the Mortar Platoon FDC, his call sign is Beefsteak 77.

2. MISSION

Patrol from our present location to vic coord_____and, report all enemy activity and/or indications of enemy movement in our AO.

3. EXECUTION

a. Departure time is 0800 hours; we will return after dark tonight.

b. We will move by squad file, "A" team leading. PFC_____

c. The route and alternate route are as indicated on this map. (Show map).

d. We will depart friendly lines through B Company area. They are aware of our departure time and will be watching for our return. Departure and return will be through the same point.

e. Remember, this is a Reconnaissance Patrol. We will attempt to avoid contact; however, if contact is made we will immediately return fire, and I will make an estimate of the situation. If contact is with a large unit, we will attempt to break contact and continue our mission. If contact is with a small unit, and I feel we are capable of eliminating it, we will maneuver using techniques of fire and movement to destroy the enemy. We will attempt to take at least one prisoner alive for questioning.

f. The Bn S-2 will debrief the entire patrol on our return.

4. ADMINISTRATION AND LOGISTICS

a. Each man will carry two rations and sufficient water.

b. PFC_____, draw an M-60 machine gun from the weapons squad. PFC_____, draw 600 rounds of M-60 ammo and act as the assistant gunner.

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c. Wounded and prisoners will be handled in accordance with the platoon SOP.

5. COMMAND AND SIGNAL

a. Signal

1) Maximum use of arm and hand signals within the patrol.
2) Current SOI is in effect. Our call sign is Beefsteak 22, NCS is Beefsteak 55 on_____Mhz. The alternate frequency is_____Mhz. Reports will be rendered to the NCS every hour, on the hour.
3) The challenge and password effective 0001 this morning till 2400 tonight is: Rusty Gate.

b. Command

1) Sgt_____is Second in command, after him Cpl_____.
2) I will be with "A" team, Sgt_____will be with "B" team.
3) The time is now_____hours, are there any question?

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APPENDIX B

ENVIRONMENTAL CONSIDERATIONS

TROPIC ENVIRONMENT CHECKLIST

a. Are all ventilating louvres provided with baffle plates and/or the louvres have reverse-curvature bends to impede the entrance of moisture?

b. Are adequate drain holes provided to drain off moisture inside closed-off areas caused by condensation dur to temperature changes when exposed to high humidity conditions?

c. Have appropriate precautions been taken to reduce the effect on test item operation due to clogging of ventilating screens or water-cooling radiators?

d. Are provisions such as shielding, insulation or warning labels provided when the temperature of surfaces with which personnel may come in contact exceeds 45°C(115°F)?

e. Is adequate protection provided on the test item to protect personnel, equipment and cargo from overhanging limbs and lianas?

f. Are provisions made on the test item to protect the operator/crew from the effects of prolonged exposure of the test item to the direct rays of the sun?

g. Is a suitable medical kit provided for on-the-site treatment against poisonous plants, insects or animals?

h. Is hermetic sealing used to keep out moisture whenever possible?

i. Is contact between corrodable metal parts and anti-fungus-treated materials avoided whenever possible?

j. Are drain holes provided, and are chassis and racks channeled to prevent moisture traps?

k. Is proper ventilation provided where personnel are performing monitoring, servicing, or other maintenance tasks?

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l. Are reflecting surfaces used on equipment which must be maintained while exposed to the sun?

m. Are materials used to the greatest extent possible for their non-corrosive and non-hygroscopic characteristics, as well as their ability to resist fungus growth?

n. Are materials used which inhibit fungus growth rather than materials which include a fungicide, or which have received surface fungistatic or fungicidal treatment?

o. Are metallic parts designed without recesses, cups or traps where liquids can accumulate, and are metals in contact chosen as close together as possible in the electromatic force series regardless of its assigned group?

p. Are exposed surfaces completely covered with some form of protective coating or surface plating?

q. Are corrosion inhibitors or preservative materials used where necessary or possible?

r. Are synthetic rubbers which are resistant to both microbiological and ozone deterioration, used wherever possible?

s. Are tarpaulins and cab tops shaped and supported to shed waste, and preclude formation of water pockets, whether the vehicle is parked or in operation?

t. To minimize moisture condensation on electronic equipment caused by high relative humidity, is the equipment kept turned on, or are lighted electric bulbs placed in or near it or are appropriate dessicants and indicators used?

u. Are the surfaces of optical elements such as lenses or mirrors kept clean of organic deposits such as finger marks, to avoid development of fungus growths on them?

v. Is daily maintenance performed on electronic equipment to keep it in an operational status?

w. Are daily checks made for fungus on the edges of insulators, keys, jacks, switches and connectors, and for the presence of small animals such as lizards and rodents?

x. Is special care and attention given to the condition of all batteries?

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y. Are precautions taken to avoid damage resulting from moisture, fungi, or insects?

z. Are electrical conductor cables and connectors kept off the ground to lessen the effects of moisture, fungi and insects?

aa. Are stenciled and decal markings protected from moisture, abrasion, corrosion and fading?

ab. Is use of sure-grip or non-slip surfacing made on all critical controls or devices?

TABLE 1

Summary of Major Environmental Effects

Environmental Factor	Principal Effects	Typical Failures Induced
High temperature	Thermal aging	Insulation failure
	Oxidation	Alteration of electrical properties
	Structural change	
	Chemical reaction	
	Softening, melting and sublimation	Structural failure
	Viscosity reduction and evaporation	Loss of lubrication properties
	Physical expansion	Structural failure
		Increased mechanical stress
		Increased wear on moving parts
High relative humidity	Moisture absorption	Swelling, rupture of container

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TABLE 1 (Continued)

		Physical breakdown
		Loss of electrical strength
	Chemical reaction	Loss of mechanical strength
	Corrosion	Interference with function
	Electrolysis	Loss of electrical properties Increased conductivity of insulators
	Dimensional Instability	Warping Surfaces
Solar Radiation	Actinic and physiochemical reactions	Surface deterioration Alteration of electrical properties
	Embrittlement	Discoloration of materials
		Ozone
Salt Spray	Chemical reactions	Increased wear
	Corrosion	Loss of mechanical strength
		Alteration of electrical properties
		Interference with function
	Electrolysis	Surface deterioration
		Structural weakening
		Increased conductivity

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TABLE 1 (Continued)

Wind	Force application	Structural collapse
		Interference with function
		Loss of mechanical strength
	Deposition of materials	Mechanical interference and clogging
		Abrasion accelerated
Rain	Physical stress	Structural collapse
	Water absorption and immersion	Increase in weight
		Aids heat removal
		Electrical failure
		Structural weakening
	Erosion	Removes protective coatings
		Structural weakening
		Surface deterioration
	Corrosion	Enhances chemical reactions
Water immersion	Corrosion of metals	Structural weakness, seizure of parts, contamination of products
	Chemical deterioration	Dissolving out and changing of materials
	High pressures (13 lbs. at 30 ft. depth)	Mechanical damage

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TABLE 1 (Continued)

Insects and bacteria	Penetration into equipment	Blockage of small parts, meters, etc.
	Nibbling by termites	Damage to plastic cables or other organic insulating materials, causing shorts.
	Spinning webs or cacoons, or depositing egg cases or droppings	Electrical shorts
Fungi	Growth of molds, hyphae	Damage to optical equipment; leakage paths in high impedance circuits; blockage of small parts, meters, etc.; breakdown of mechanical strength of all organic materials
		Corrosion of metals due to metabolic products
Ozone	Chemical reactions:	Rapid oxidation
	Crazing, cracking	Alteration of electrical properties
	Embrittlement	Loss of mechanical strength
	Granulation	Interference with function
	Reduction dielectric strength of air	Insulation breakdown and arcover

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